

KILPISJARVI SAANA, Finland

WMO: 027010

Lat: 69.043N

Lon: 20.851E

Elev: 3310

StdP: 13.02

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-3.7	-12.9	3.1	-7.6	-7.9	4.1	-2.0	57.1	13.7	52.6	11.5	25.0	140	1.117

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	64.7	54.6	60.9	53.2	57.0	50.7	56.2	62.7	54.1	58.9	51.9	55.9	14.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.7	69.4	56.9	51.7	64.4	55.3	49.7	59.7	53.9	25.3	62.6	23.9	59.3	22.5	55.6	63.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 48.1	42.6	37.8	DB	-13.8	69.2	6.3	4.8	-18.3	72.7	-21.9	75.5	-25.4	78.1	-30.0	81.6	(4)
(5)			WB	-12.7	58.5	5.2	3.0	-16.5	60.7	-19.5	62.4	-22.5	64.1	-26.3	66.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	11.6	12.3	15.0	21.8	30.8	39.5	48.2	45.2	37.9	26.4	19.7	15.6	(6)
(7)		DBStd	14.41	8.17	8.32	6.23	6.51	8.39	7.69	7.60	6.45	6.46	6.39	6.62	6.43	(7)
(8)		HDD50	8470	1189	1054	1084	846	598	327	127	177	362	732	909	1065	(8)
(9)		HDD65	13830	1654	1474	1549	1296	1060	766	520	613	812	1197	1359	1530	(9)
(10)		CDD50	117	0	0	0	0	3	12	72	29	1	0	0	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH74	2	0	0	0	0	0	0	2	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	17.8	20.0	19.8	19.9	16.4	16.7	17.0	15.4	15.5	17.8	17.5	17.9	19.6	(14)
(15)	Precipitation	PrecAvg	19.6	2.0	1.5	1.3	0.9	1.1	1.7	2.7	1.8	1.4	1.6	1.4	2.0	(15)
(16)		PrecMax	24.3	5.9	3.5	6.3	3.5	2.0	3.3	5.2	3.9	2.5	3.1	3.1	4.1	(16)
(17)		PrecMin	14.0	0.7	0.3	0.3	0.2	0.3	0.2	0.7	0.8	0.4	0.2	0.1	0.7	(17)
(18)		PrecStd	2.4	1.2	0.7	1.4	0.6	0.5	0.7	1.1	0.7	0.7	0.8	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.6	30.3	42.8	60.8	64.3	72.5	65.5	56.3	45.3	41.3	32.1	(19)
(20)			MCWB	30.6	31.5	29.5	36.7	48.2	53.5	58.2	55.9	44.4	36.3	36.1	32.1	(20)
(21)		2%	DB	27.9	27.7	27.8	37.6	51.9	58.2	67.2	61.5	51.9	40.5	34.9	28.8	(21)
(22)			MCWB	27.5	27.0	27.3	34.3	45.1	50.7	55.7	53.6	45.4	37.7	34.1	28.6	(22)
(23)		5%	DB	25.2	25.7	25.7	32.4	46.8	53.9	63.3	57.6	48.8	38.0	31.2	26.3	(23)
(24)			MCWB	24.8	24.7	25.3	31.1	41.0	47.5	54.4	51.7	44.7	36.5	30.9	26.0	(24)
(25)		10%	DB	22.8	23.2	23.7	30.2	42.4	51.1	59.7	54.2	46.3	35.4	28.3	23.9	(25)
(26)			MCWB	22.2	22.1	23.3	29.4	38.1	46.4	52.6	50.1	43.5	34.1	28.0	23.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	31.6	29.6	38.5	48.9	56.2	60.3	60.0	49.8	41.3	37.3	32.2	(27)
(28)			MCDWB	31.0	31.8	29.6	41.9	57.4	61.6	67.9	63.1	51.8	42.0	39.6	32.3	(28)
(29)		2%	WB	27.9	27.4	27.3	34.8	45.4	52.1	57.1	54.8	47.0	38.3	33.5	29.0	(29)
(30)			MCDWB	28.4	27.7	27.7	37.7	51.6	55.0	65.4	58.8	49.5	39.1	34.9	29.0	(30)
(31)		5%	WB	25.2	25.3	25.1	31.8	41.4	49.8	55.3	52.5	45.3	36.4	30.6	26.3	(31)
(32)			MCDWB	25.5	25.6	25.4	32.7	46.5	53.9	61.9	56.3	47.6	37.6	31.0	26.5	(32)
(33)		10%	WB	22.7	22.6	23.2	29.6	38.4	46.4	53.4	50.5	43.7	34.1	27.8	23.5	(33)
(34)			MCDWB	23.1	23.0	23.5	30.4	42.0	50.5	58.2	54.3	45.9	35.5	28.1	23.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	8.0	7.4	6.9	8.5	9.9	10.6	9.0	6.9	5.7	6.3	6.5	(35)
(36)			MCDBR	9.4	10.6	10.3	8.9	13.8	14.6	14.6	12.8	10.6	9.1	8.5	8.4	(36)
(37)			MCWBR	9.3	10.8	10.8	7.3	8.2	8.7	6.9	7.6	7.3	7.2	7.5	8.7	(37)
(38)		5% WB	MCDWB	9.3	11.3	10.2	8.6	13.7	13.2	12.8	11.1	9.1	8.0	8.4	8.6	(38)
(39)			MCWBR	9.3	11.5	10.4	7.3	8.8	8.8	6.9	7.9	7.0	7.2	7.7	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.162	0.212	0.256	0.278	0.293	0.299	0.304	0.292	0.266	0.224	0.163	N/A	(40)	
(41)		taud	1.906	2.095	2.207	2.251	2.350	2.524	2.542	2.554	2.579	2.245	1.951	N/A	(41)	
(42)		Ebn at Noon	21	183	247	276	285	287	281	273	253	179	17	0	(42)	
(43)		Edh at Noon	2	13	23	30	31	27	26	23	17	11	1	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6	123	563	1189	1588	1565	1446	1026	547	170	22	0	(44)	
(45)		RadStd	1	14	33	63	128	126	127	78	50	20	2	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page